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An Innovative Fluid Technology and Advanced Digital Workflow Set a New Standard for Carbonate Matrix Stimulation in High-Temperature Extended-Reach Wells: An Exceeding Expectations Story from the UAE

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Abstract

Designing acid stimulation in extended-reach wells is challenging. This study adopts a digital workflow that integrates data from previous treatments, lab experiments, and production performance to design and optimize acid treatments in carbonate formations. The wells under study are horizontal and completed as open holes. The laterals are over 13,000 ft. long, and the average formation permeability is 4 mD. The formation temperature is 270°F.

An advanced acid placement simulator and production analysis software evaluated a previous treatment involving emulsified acid in the same field. The evaluation of the previous treatment was a blind test where only pre-stimulation production data and acid treatment data were provided. The acid placement simulator used the inputs from the operator as an initial guess to match actual treatment pressures. Measured wellhead pressures and bottomhole pressures were used to adjust the fluid's friction multiplier. In agreement with the operator, initial skin and some other parameters were modified to achieve an accurate pressure match. The post-stimulation skin from the simulator was then used in production analysis software to predict the post-stimulation drawdown. The reduction in actual drawdown before and after the treatment, as well as the simulated reduction, was found to be within 10%. This demonstrates the accuracy of the acid placement simulator in modeling the carbonate acidizing process in long laterals.

Core flow experiments were conducted to evaluate the performance of emulsified acid and polymer-free delayed acid on outcrop and reservoir cores. The experimental data were utilized to fine-tune the acid performance curves in the simulator. A validated upscaling scheme was implemented to compare the performance of the two acids at the field scale. The simulations showcased that polymer-free delayed acid consistently outperformed emulsified acid, in terms of placement, friction, and acid penetration in this specific application.

The design for the well treatments involved conducting numerous runs on the acid placement simulator. The runs were ranked based on acid coverage and skin distribution along the lateral. The post-stimulation skin distributions were then input into the production analysis software to predict drawdown and

improvements in PI. Treatment decisions were based on optimum dosage rates/economics, PI without compromising target skin reductions. The results from eight stimulated wells reveal that predicted pressures from the simulator closely match gauge pressures.

This integrated digital workflow facilitates accurate engineered solutions and economics-driven optimization of carbonate acidizing stimulation. It also highlights how full engagement/collaboration between the operator and service company enhances performance and optimum design. In this stimulation campaign, continuous integration and engagement translated to a 30% reduction in acid volume, a 65% increase in injection rate, a 100% improvement in post-stimulation PI, and 560 metric tons less CO₂ emitted compared to previous well treatments in the same field.

Introduction

The United Arab Emirates (UAE) continues to advance hydrocarbon recovery through the deployment of cutting-edge stimulation technologies in geologically complex carbonate reservoirs. These formations, typically characterized by low permeability, heterogeneous porosity, and high concentrations of sour gas, pose significant challenges for matrix stimulation—particularly in high-temperature, open-hole extended-reach wells (ERWs). Conventional stimulation fluids and workflows often fail to deliver consistent results under such demanding conditions, leading to inefficient acid placement, limited reservoir contact, and suboptimal post-treatment productivity.

The reservoir discussed in this study is located onshore in Abu Dhabi, near the border with Saudi Arabia. It is a dolomitic carbonate formation producing ultra-sour lean gas, with hydrogen sulfide (H₂S) concentrations exceeding 20% and carbon dioxide (CO₂) levels between 8–10%. The highly corrosive nature of the reservoir fluids renders lower completion installations prohibitively expensive, necessitating open-hole completions across reservoir sections exceeding 13,000 ft. Compounding the challenge is a bottomhole static temperature of approximately 270°F, which significantly increases the reactivity of conventional hydrochloric acid (HCl) systems, often resulting in rapid acid spending near the wellbore and ineffective stimulation deeper into the formation.

To overcome these limitations, a novel stimulation strategy was developed following an extensive fluid screening and design process. A single-phase, polymer-free, retarded acid system (Single-Phase PFDA) was selected for its thermal stability and delayed reactivity, enabling high-rate bullheading treatments without compromising acid integrity or placement efficiency. This fluid technology was integrated with an advanced digital workflow that leveraged real-time data analytics, predictive modeling, and remote execution capabilities to optimize treatment design and execution dynamically.

This paper presents the engineering rationale, deployment strategy, and field results of this integrated stimulation solution. The approach not only addressed the unique challenges of ultra-sour, high-temperature carbonate reservoirs but also established a new benchmark for stimulation design, stimulation performance, operational efficiency, and digital enablement in extended-reach well environments.

Evaluation and Selection Process

Innovative Fluid Technology

The stimulation of ultra-sour, high-temperature carbonate reservoirs requires a fluid system capable of maintaining chemical integrity under extreme downhole conditions while delivering controlled reactivity to ensure deep matrix penetration. Conventional hydrochloric acid (HCl) systems exhibit rapid reaction kinetics at elevated temperatures, particularly in dolomitic formations, resulting in near-wellbore acid spending and ineffective stimulation across extended reservoir sections.

To overcome these limitations, a single-phase, polymer-free, retarded acid system (Single-Phase PFDA) was selected. This formulation utilizes proprietary retardation chemistry to modulate hydrogen ion

availability, slowing the acid-rock reaction rate and enabling deeper penetration into the dolomitic matrix. The absence of polymers eliminates the risk of formation damage, which is particularly critical in gas-producing environments.

The polymer-free, single-phase delayed acid fluid technology is applicable for carbonate formations as it is based on HCl acid that has high dissolving power to the calcareous materials and effectively increase the permeability and productivity of the reservoir leading to optimized hydrocarbon recovery.

Simpler and safer. The new acid fluid technology is a single-phase fluid and doesn't contain emulsion neither polymer gels, thus simplifying operations, exhibiting friction and viscosity properties similar to regular HCl for high pumping rates and minimizing surface horsepower and footprint requirements. Because it is a single-phase system, there are no special mixing requirements or need for diesel emulsion, improving HSE, and minimizing carbon footprint. The acid fluid system can also be mixed on-the-fly, further reducing downtime. The new acid fluid technology effectively provides zonal coverage with deep reservoir contact and deployment flexibility through bull heading or through coiled tubing.

Extended. Regular HCl reacts rapidly with the formation, resulting in face dissolution and near-wellbore spending that limit the success of the acid stimulation treatment. The polymer-free, single-phase acid fluid technology is 100 times more retarded at 275°F compared to the regular HCl which allows for deeper matrix penetration of live acid, creation of single dominant wormholing reservoir contact and significant skin reduction post-acid stimulation job.

Cleaner. The polymer-free acid fluid technology is a single-phase fluid that doesn't contain any emulsions or polymer gels. The use of the new acid fluid technology eliminates the risk of any emulsions that could be retained in the formation to cause damage or water wetting the rock as in the case of the emulsified acid and leaves the formation free from any polymer gels residue as in the case of gelled acid to maximize the stimulation treatment success.

Laboratory Evaluation

An operator driven comprehensive laboratory testing program was conducted to validate the fluid's performance under reservoir-representative conditions:

- **Acid Corrosion Testing:** Conducted at 300°F for 5 hours using Alloy 825 coupons exposed to 1% H₂S. Two acid concentrations (15% and 20%) were tested with corrosion inhibitors (CI-42) and intensifiers (HyTemp-O). Both systems passed with corrosion rates of 0.0179 and 0.0252 lb/ft² respectively, and minimal pitting observed.

Table 1—Corrosion Test Results

Test	Acid	Temp	Exposure Time	Metal	Corrosion Inhibitor	Corrosion Intensifier	Corrosion Rate (lb/sq.ft)	Pitting	Pass/Fail
1	20% Polymer-Free Delayed Acid System	300°F	5 Hrs.	Alloy 825	20 gpt CI-42	20 gpt HyTemp-O	0.0252	1	Pass
2	15% Polymer-Free Delayed Acid System	300°F	5 Hrs.	Alloy 825	15 gpt CI42	15 gpt HyTemp-O	0.0179	0-1	Pass

Table 2—Corrosion Test - Coupon Results, Sample #1

Test #1	
Initial	After 5 hours exposure
	

Table 3—Corrosion Test - Coupon Results, Sample #2

Test #2	
Initial	After 5 hours exposure
	

- Coreflow Testing:** Indiana limestone cores (20 in) were used to evaluate acid performance at 300°F and 5 cc/min flow rate. Pore volume to breakthrough (PVBT) values of 0.41 and 0.32 were recorded for 15% and 20% Single-Phase PFDA respectively, indicating efficient wormhole propagation.

Table 4—Pore Volume Breakthrough Test Results

Test Number	Core Type	Core Length	Flow Rate	Acid	PVBT Results
1	Indiana Limestone	20 in	5 cc/min	15% Single-Phase PFDA	0.41
2	Indiana Limestone	20 in	5 cc/min	20% Single-Phase PFDA	0.32

Post-coreflow CT scans confirmed the formation of well-defined wormholes, validating the acid's ability to propagate effectively through the matrix under simulated reservoir conditions.

Table 5—CT Scan Results post PVBT - wormhole created upon Single Phase retarded acid injected at 5cc/min

Chart 1 – Core Flow Analysis at 300°F	
Chart 2 – Core Flow Analysis at 300°F	

- **Static Dissolution Test:** Conducted at ambient and 160°F using CB-107 cores. Dissolution remained well below the industry threshold of 10% by weight, confirming controlled reactivity.

Table 6—Static dissolution test result

Temperature	Acid	% Solubility
Room Temp	20% Single-Phase PFDA	4.60%
Room Temp	15% Single-Phase PFDA	6.00%
160°F	20% Single-Phase PFDA	9.25%
160°F	15% Single-Phase PFDA	8.50%

These results confirm the fluid's suitability for high-rate bullheading in open-hole completions exceeding 13,000 ft, where mechanical diversion is impractical. The system's thermal stability, controlled reactivity, viscosity profile and sour gas compatibility make it a robust solution for carbonate matrix stimulation in the UAE's most challenging reservoir environments.

It is also critical to note that these tests results achieved are consistently repeatable (low margin of deviation) indicating a level of repeatability confidence that's commonly not witnessed on other systems.

Advance Digital Workflow

Fluid Placement Simulator. Accurate prediction and analysis of wormhole development and the resulting skin effect depend on two key factors: (1) the ability to estimate reservoir-face pressure, and (2) a thorough understanding of the physics and chemistry governing acid flow in porous media. In many cases, reservoir-face pressure must be inferred from surface treating pressure. Occasionally, downhole gauges placed at the bottom of the upper completion provide direct measurements. However, in horizontal wells, the reservoir face may be significantly distant from these gauges, necessitating the use of detailed mathematical models for reliable pressure estimation (Furui et al., 2012).

As treatment fluids travel from the surface through the wellbore, they experience pressure losses due to friction along the tubular walls and pressure gains from changes in vertical depth (Ali et al. 2024). Flow within the horizontal section is primarily governed by frictional forces. In the lower completion, fluids typically enter the reservoir radially through limited entry points. The simplest configuration—an openhole completion—allows radial flow without mechanical restrictions. More advanced completions, however, require fluids to pass through orifices or devices such as Inflow Control Devices (ICDs), introducing additional frictional losses that must be accounted for using mathematical models (Mogensen et al., 2020).

In some scenarios, a single ICD may be used to stimulate over 200 feet of reservoir, demanding sophisticated algorithms to model post-ICD flow and ensure precise fluid distribution. Once the fluid reaches the reservoir face at the correct pressure, classical production engineering equations can be applied to calculate the injection rate for each zone.

Acid treatments, commonly used in carbonate formations, aim to create narrow channels known as wormholes that bypass formation damage and enhance well productivity or injectivity. A validated model (Ali, 2022) predicts wormhole growth based on variables such as injection rate, acid type and concentration, temperature, rock type, and mineralogy. The resulting wormholes contribute to reduced skin and improved well performance.

Prediction of wormhole growth with PFDA system. The Ali (2022) model, represented in Equation 1, estimates wormhole growth rates across different zones. This model incorporates key parameters such as velocity at the formation face, wellbore or openhole diameter, formation characteristics, acid type and concentration, and temperature to accurately determine the wormhole tip location.

$$V_{wh} = W_{eff} * (V)^{e_1} * (1 - \exp(-W_b * (V)^{e_2}))^{e_3} \quad \text{Eq. 1}$$

Where, V_{wh} is the wormhole velocity, V is Darcy velocity at the formation face, W_{eff} is the wormhole efficiency factor, W_b is the wormhole B-factor and e_1 - e_3 are tuning parameters.

W_{eff} and W_b are functions of acid concentration, diffusion coefficient, wormhole length, formation face area, mineralogy, acid formulation, and permeability.

This model has been validated against more than 600 linear core flooding experiments. Fig. 1 demonstrates the model's capability to predict the impact of temperature on the pore volume of acid required to achieve breakthrough (PVBT) for various HCl concentrations. The results show strong agreement between model predictions and experimental data, confirming its accuracy in capturing temperature effects.

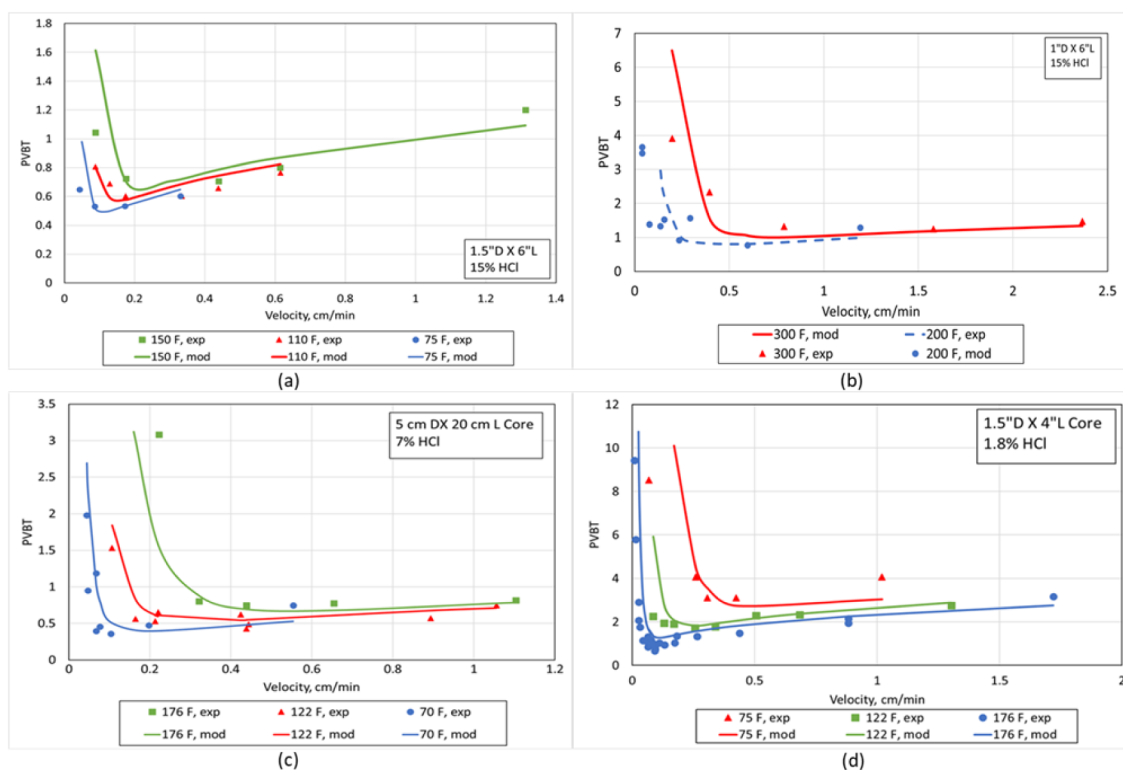


Figure 1—(a) Effect of temperature between 75°F to 150°F on acid flow using 15% HCl. (b) Effect of temperature between 200°F to 300°F on acid flow using 15% HCl. (c) Effect of temperature from 70°F to 176°F on acid flow using 7% HCl. (d) Effect of temperature from 75°F to 176°F on acid flow using 1.8% HCl. (After Ali et al. 2024)

Experimental results have demonstrated that PFDA outperforms emulsified acid at elevated temperatures (Zakaria et al., 2022). Two sets of experiments using 28% HCl-based PFDA were conducted at temperatures of 225°F and 300°F, utilizing Indiana limestone cores with diameters of 6 inches and 20 inches, respectively. Figure 3 compares the experimental data with model predictions, showing that the combined increase in core length and temperature has only a minimal impact on acid performance. Also, experiments have been conducted on dolomite cores to study the effectiveness of PFDA in treating dolomite formations. 1.5 in. diameter × 6 in. length cores were treated with 20% HCl based PFDA at 250°F. Fig. 14 shows a comparison between experiments and model predictions.

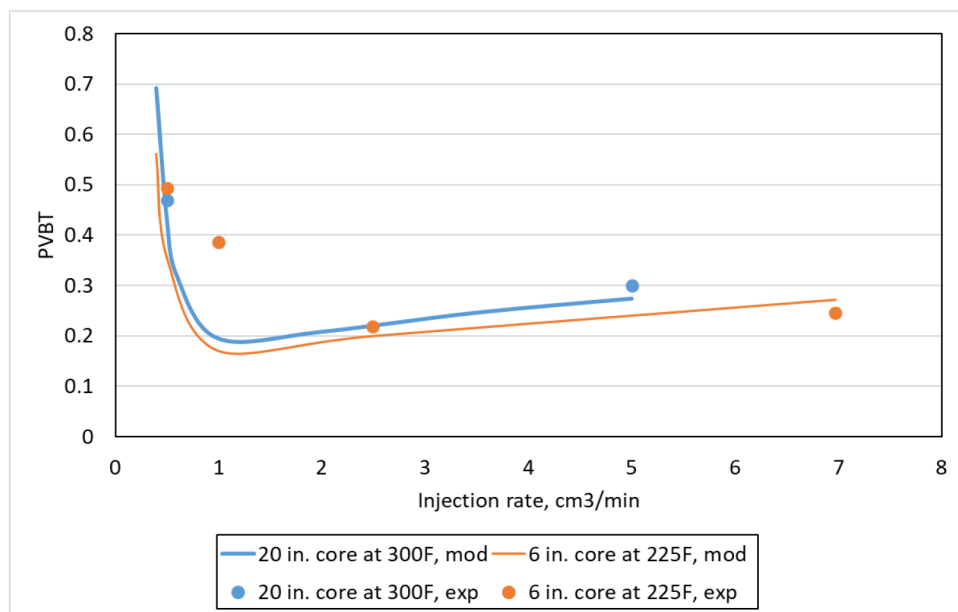


Figure 2—Flow of PFDA in limestone cores. Symbols are experiments. Lines are model predictions (After Ali and Zakaria 2023)

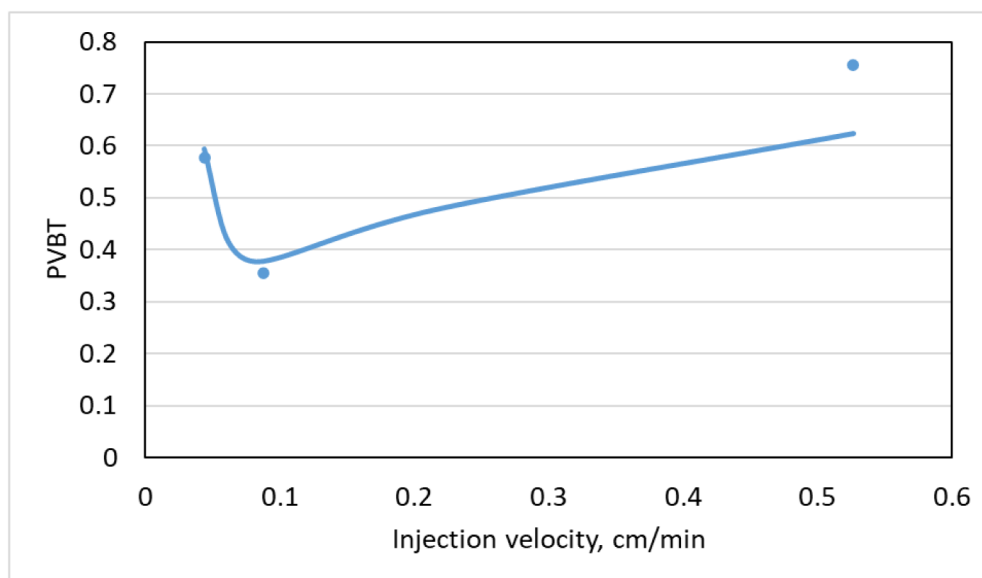


Figure 3—Flow of PFDA in dolomite cores. Symbols are experiments. Lines are model predictions (After Ali and Zakaria 2023)

The performance of PFDA under radial conditions is shown in Fig.4. It shows that pumping a very low dosage (20 gal/ft.) of PFDA can effectively lower the skin to less than -3 .

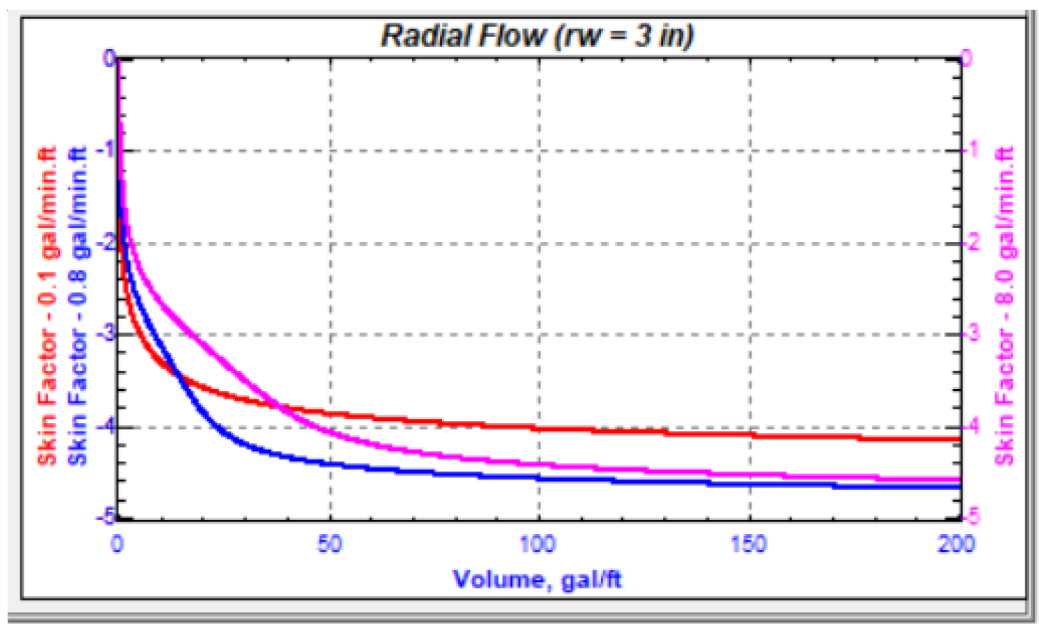


Figure 4—Skin evolution with acid volume at three injection rates. Limestone treated with 28% HCl based PFDA at 225°F

Predictive modeling and decision support. Stimulating ultra-sour, high-temperature carbonate reservoirs in extended-reach open-hole wells demands a digitally enabled workflow that supports dynamic decision-making, real-time monitoring, and precise execution. In this project, a fully integrated digital stimulation workflow was deployed to complement the advanced fluid system, ensuring optimal treatment design and execution across reservoir sections exceeding 13,000 ft.

A key component of this workflow was the use of predictive modeling, which incorporated laboratory coreflow results—particularly pore volume to breakthrough (PVBT)—to fine-tune acid performance across long lateral sections. These models were calibrated using validated skin values and acid penetration profiles relative to reservoir permeability, enabling accurate simulation of stimulation behavior under field conditions.

Pre and Post Stimulation Analysis. The workflow integrated pre-stimulation reservoir flow data to establish baseline productivity. Post-stimulation flow rates were then compared to assess the effectiveness of acid placement and stimulation coverage. This comparative analysis confirmed acid distribution along the extended open-hole lateral and validated the treatment design.

External software tools were used to estimate well productivity, match initial skin values, and simulate post-stimulation performance. The acidizing model assumed a pre-stimulation skin of +11 and a damage depth of 3 ft across the lateral.

Well Introduction Information

Utilization of external software to estimate the well productivity as Gas Rate, Gas productivity index, and matching initial skin from pre-stimulation.

Parameter	Value
Completion Date	October 2023
Tubing	7" x 5.5"
Horizontal Length	11,869 ft
Reservoir Pressure	5,000 psi
Pre-Acid Gas Rate	83 MMSCFD
FWHP	2,615 psi
FWHT	206°F
Stimulation Volume	4,580 bbls
Acid Volume per Foot	15 gal/ft

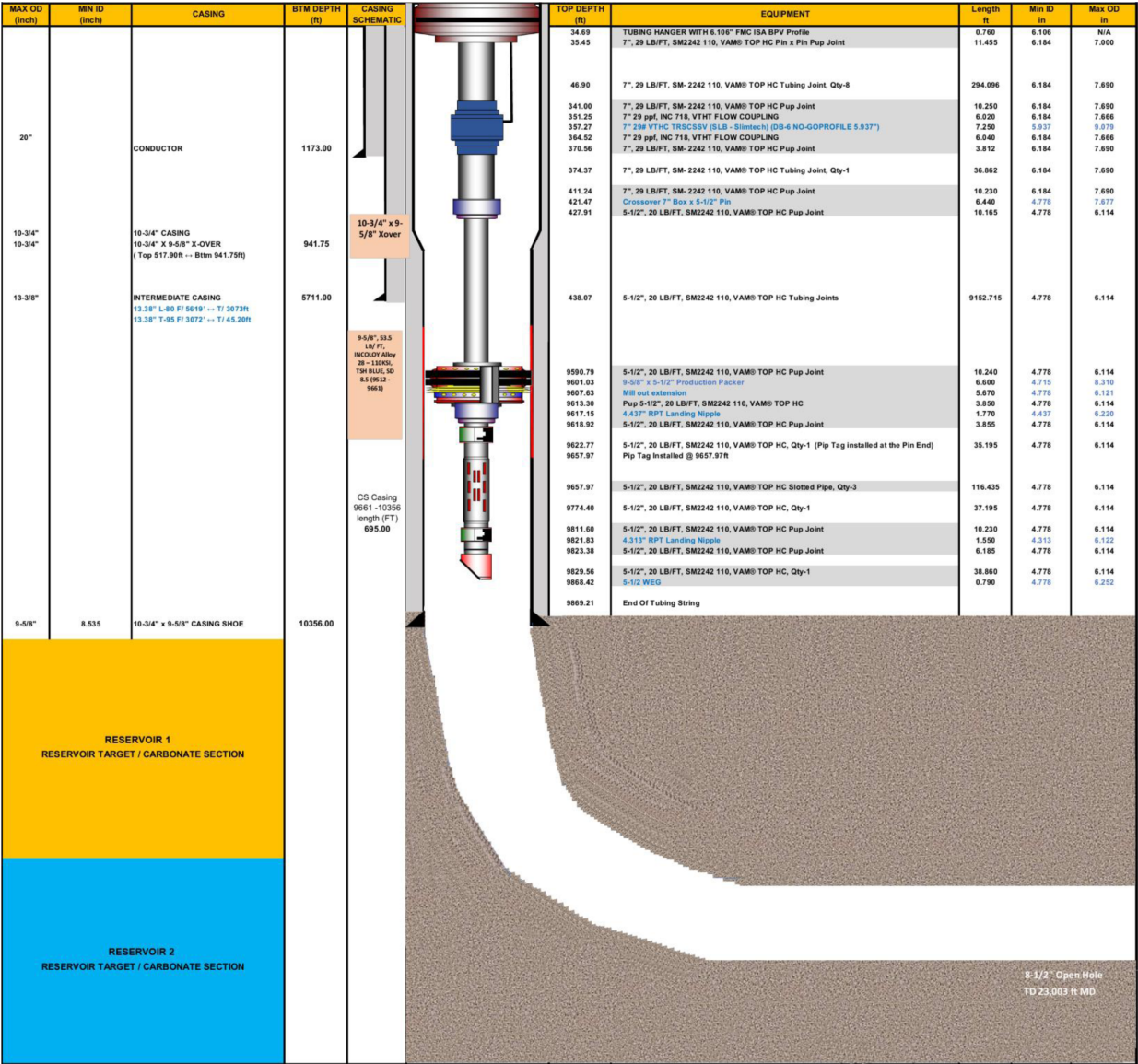


Figure 5—Wellbore Completion Schematic

Pre-Stimulation Performance.

Table 7—Pre-Stim Review - External Software

Calculated BHP psi	Measured BHP psi	Gas rate (MMSCFD)	THP psi	Gas .P.I
4367.43	4371	82.93	2615	0.127818

Post-Stimulation Performance.

Table 8—Post Stim Review - External Software

Calculated BHP psi	Measured BHP psi	Gas rate (MMSCFD)	Gas .P.I
4821.75	4793	89	0.4594

Property Digital Software: Acidizing Model. Proprietary software was used to predict the treating pressure, flow rates and skin post stimulation, well information were input, and common considerations were assumed as initial pre stimulation reservoir skin across lateral as +11 and depth of damage of 3 ft.

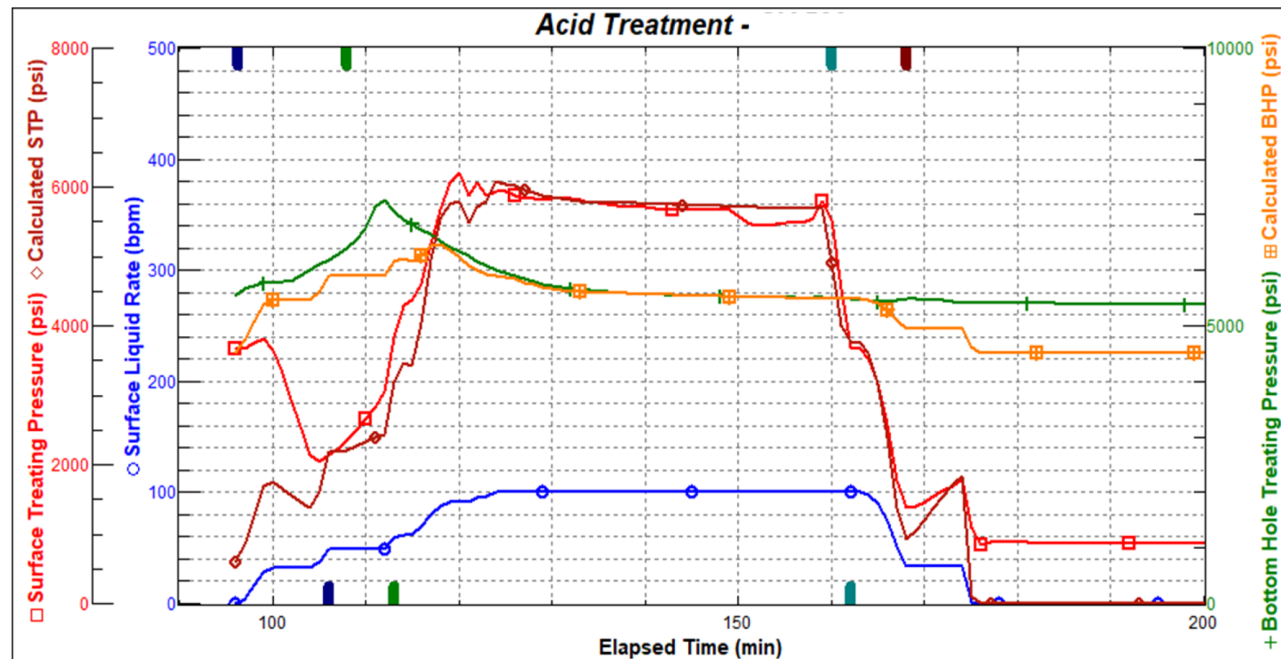


Figure 6—Stimulation Review (Post Acid Stimulation) - Calibrated model proprietary Software

This advanced digital workflow enabled precise acid placement, minimized operational risk, and enhanced reservoir contact. The integration of laboratory insights, predictive modeling, and field performance data sets a new benchmark for carbonate matrix stimulation in extended-reach, high-temperature sour gas environments.

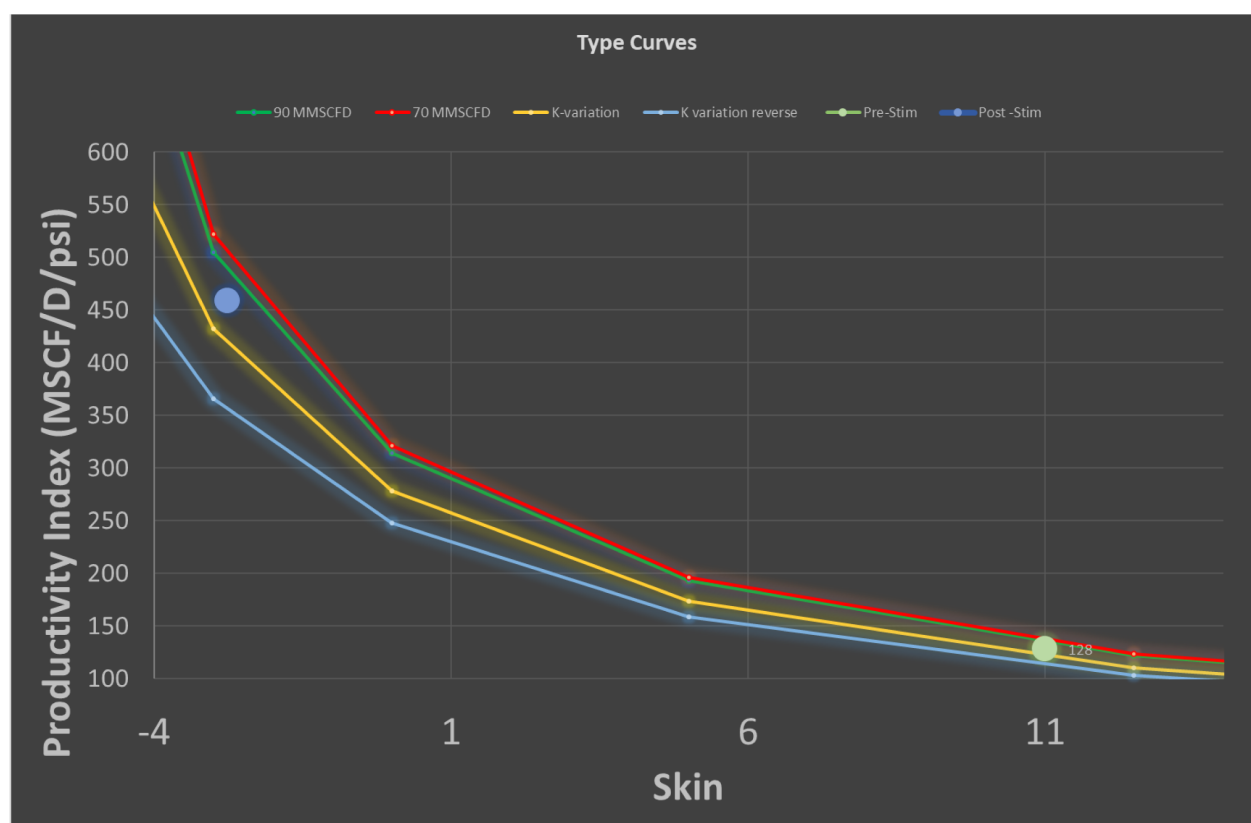


Figure 7—Productivity Index Pre and Post Stimulation vs initial and post Skin

Conclusions

This paper presents an innovative approach that integrates advanced digital workflows and innovative acid fluid technology along with stimulation engineering, unlocking the full productivity potential. By utilizing a real-time acid placement simulator, we can accurately predict wormhole growth and skin profiles. Furthermore, the combination with production analysis software facilitates dynamic treatment optimization, ensuring both effective acid coverage and cost efficiency. The strong correlation observed between simulated and actual drawdown values, within a margin of just 10%, attests to the reliability of these predictive models. This method establishes a new standard for data-driven stimulation design in complex carbonate reservoirs.

The integrated stimulation campaign achieved substantial operational and environmental benefits. Through collaborative design and execution between the Company and Service Provider, the optimized treatments led to a 30% reduction in acid volume, a 65% increase in injection rate, and a 100% improvement in post-stimulation productivity index. Additionally, the campaign reduced CO₂ emissions by 560 metric tons compared to previous treatments. These outcomes underscore the value of combining innovative fluid systems with digital enablement to drive sustainable and high-performance reservoir stimulation.

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